

Limo Gazebo Humble Guidelines



Status

Not started

0. Package Contents

- README.pdf
- docker.sh
- limo_gazebo_humble.zip

1. Local에서 도커 환경 구축

- ubuntu 버전 확인

```
lsb_release -a
```

```
park@park:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:   Ubuntu 22.04.5 LTS
Release:      22.04
Codename:     jammy
```

- docker image pull

```
docker pull yspark98/limo:gazebo_humble
```

```

park@park:~$ docker pull yspark98/limo:gazebo_humble
gazebo_humble: Pulling from yspark98/limo
d544298cabd5: Pull complete
4d311ecb7e58: Pull complete
f3df817f7955: Pull complete
e782053ab619: Pull complete
2047a400df95: Pull complete
de282407801b: Pull complete
05637bbf7e18: Pull complete
2dbefbfa038f7: Pull complete
6ffda27a0e7d: Pull complete
43e85ec64126: Pull complete
d67e09d14c3f: Pull complete
e88045bc6efa: Pull complete
77fb41fec40e: Pull complete
Digest: sha256:ac08ac98b81f06e077bfbdae90e9f99df6ea2ea4db8b42cb0411e5f0d9b8bece6
Status: Downloaded newer image for yspark98/limo:gazebo_humble
docker.io/yspark98/limo:gazebo_humble

```

- 확인

```
docker images
```

```

park@park:~$ docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
yspark98/limo       gazebo_humble      a8348fee1e08       43 minutes ago     4.14GB

```

- 내장 그래픽 확인

```
ls -l /dev/dri
```

```

park@park:~$ ls -l /dev/dri
total 0
drwxr-xr-x  2 root root      80 9월  8 11:58 by-path
crw-rw----+ 1 root video 226,   1 9월  8 11:58 card1
crw-rw----+ 1 root render 226, 128 9월  8 11:58 renderD128

```

- limo_gazebo_humble.zip unzip

```
unzip ws_limo_humble.zip
```

```

park@park:~/Documents$ ls
AnyDesk ws_limo_humble.zip ws_limo_melodic ws_ros ws_sdk ws_topst ws_view_height
park@park:~/Documents$ unzip ws_limo_humble.zip
Archive: ws_limo_humble.zip
  creating: ws_limo_humble/
  creating: ws_limo_humble/install/
  inflating: ws_limo_humble/install/setup.bash
  inflating: ws_limo_humble/install/_local_setup_util_sh.py
  inflating: ws_limo_humble/install/setup.ps1

```

- docker.sh → limo_gazebo_humble 내부로 이동

```

$ docker.sh
1  #!/bin/bash
2
3  xhost +SI:localuser:root
4
5  docker run -it \
6    --name limo_gazebo_humble \
7    --network host \
8    --ipc host \
9    --device /dev/dri:/dev/dri \
10   --workdir {your/workspace} \
11   -e DISPLAY="$DISPLAY" \
12   -e QT_X11_NO_MITSHM=1 \
13   -v /tmp/.X11-unix:/tmp/.X11-unix:rw \
14   -v /etc/localtime:/etc/localtime:ro \
15   -v {your/workspace}:{your/workspace}:rw \
16   yspark98/limo:gazebo_humble

```

{your/workspace} → limo_gazebo_humble 실제 경로로 수정

ex)

```

park@park:~/Documents/ws_limo_humble$ ls
build docker.sh install log roboracer_map src
park@park:~/Documents/ws_limo_humble$ pwd
/home/park/Documents/ws_limo_humble
park@park:~/Documents/ws_limo_humble$

```

```
$ docker.sh
1  #!/bin/bash
2
3  xhost +SI:localuser:root
4
5  docker run -it \
6      --name limo_gazebo_humble \
7      --network host \
8      --ipc host \
9      --device /dev/dri:/dev/dri \
10     --workdir /home/park/Documents/ws_limo_humble \
11     -e DISPLAY="$DISPLAY" \
12     -e QT_X11_NO_MITSHM=1 \
13     -v /tmp/.X11-unix:/tmp/.X11-unix:rw \
14     -v /etc/localtime:/etc/localtime:ro \
15     -v /home/park/Documents/ws_limo_humble:/home/park/Documents/ws_limo_humble:rw \
16     yspark98/limo:gazebo_humble
```

- `docker.sh` 권한 부여

```
sudo chmod +x docker.sh
```

```
park@park:~/Documents/ws_limo_humble$ sudo chmod +x docker.sh
[sudo] password for park:
```

- docker 실행

```
./docker.sh
```

```
park@park:~/Documents/ws_limo_humble$ ./docker.sh
localuser:root being added to access control list
root@park:/home/park/Documents/ws_limo_humble#
```

- 도커 컨테이너 내부에서 내장 그래픽 확인

```
ls -l /dev/dri
```

```
root@park:/home/park/Documents/ws_limo_humble# ls -l /dev/dri
total 0
crw-rw---- 1 root video 226, 1 Sep 8 15:48 card1
crw-rw---- 1 root 110 226, 128 Sep 8 15:48 renderD128
```

2. Gazebo 실행

- 빌드

```
source /opt/ros/humble/setup.bash
```

```
colcon build --symlink-install
```

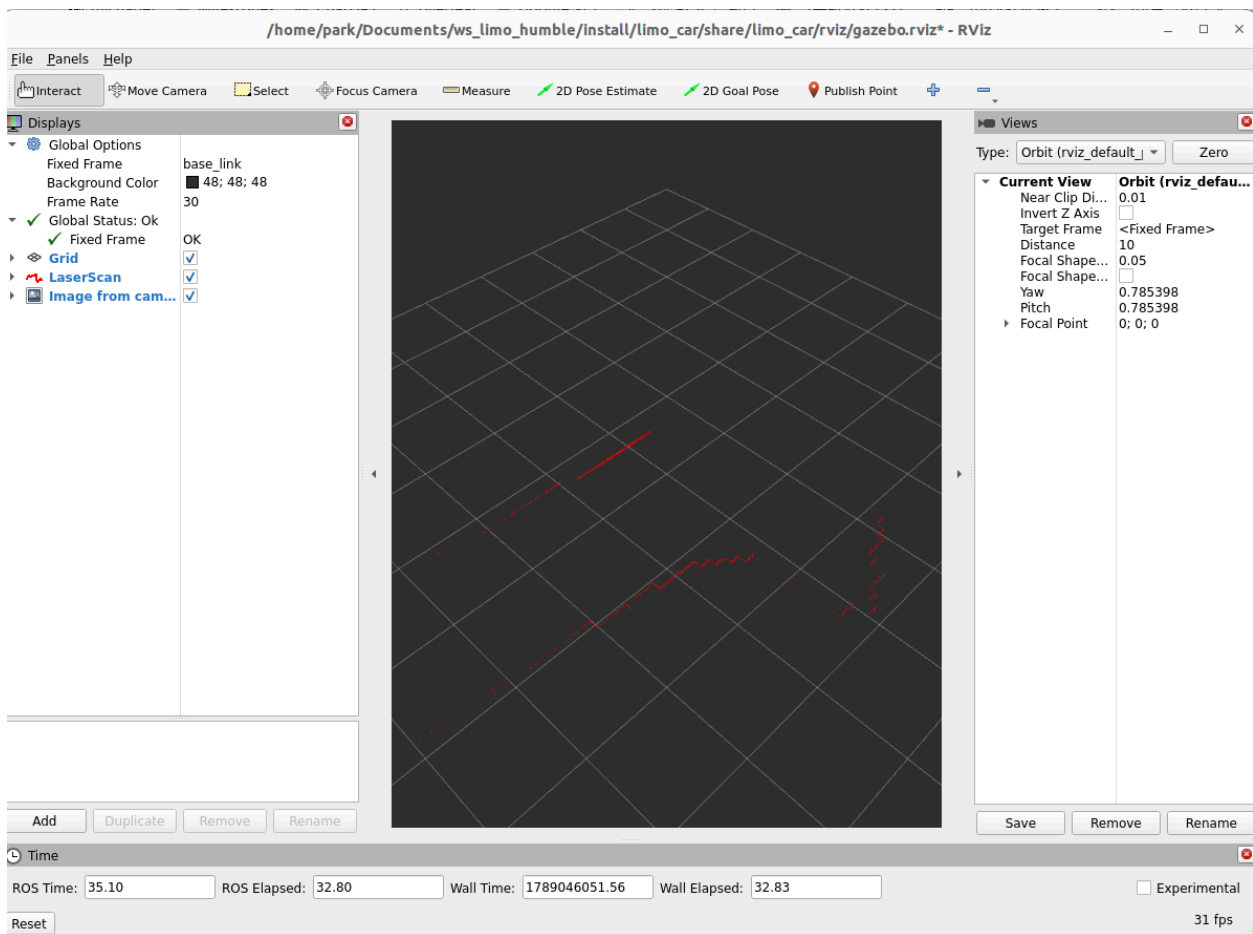
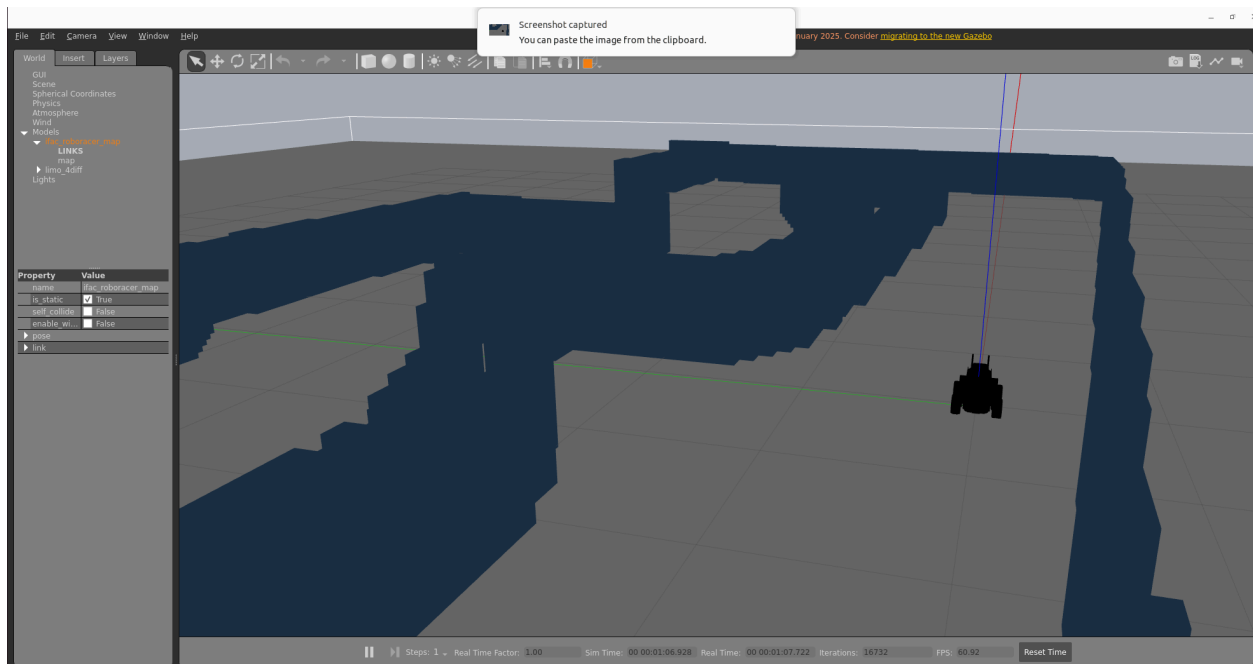
```
source install/setup.bash
```

```
root@park:/home/park/Documents/ws_limo_humble# source /opt/ros/humble/setup.bash
root@park:/home/park/Documents/ws_limo_humble# colcon build --symlink-install
Starting >>> limo_msgs
[0.191s] WARNING:colcon.colcon_ros.prefix_path.ament:The path '/home/park/Documents/ws_limo_humble/install/limo_base' in the environment variable AMENT_PREFIX_PATH doesn't exist
[0.191s] WARNING:colcon.colcon_ros.prefix_path.ament:The path '/home/park/Documents/ws_limo_humble/install/limo_msgs' in the environment variable AMENT_PREFIX_PATH doesn't exist
[0.191s] WARNING:colcon.colcon_ros.prefix_path.ament:The path '/home/park/Documents/ws_limo_humble/install/limo_description' in the environment variable AMENT_PREFIX_PATH doesn't exist
[0.191s] WARNING:colcon.colcon_ros.prefix_path.ament:The path '/home/park/Documents/ws_limo_humble/install/limo_car' in the environment variable AMENT_PREFIX_PATH doesn't exist
[0.191s] WARNING:colcon.colcon_ros.prefix_path.catkin:The path '/home/park/Documents/ws_limo_humble/install/limo_base' in the environment variable CMAKE_PREFIX_PATH doesn't exist
[0.191s] WARNING:colcon.colcon_ros.prefix_path.catkin:The path '/home/park/Documents/ws_limo_humble/install/limo_msgs' in the environment variable CMAKE_PREFIX_PATH doesn't exist
[0.191s] WARNING:colcon.colcon_ros.prefix_path.catkin:The path '/home/park/Documents/ws_limo_humble/install/limo_description' in the environment variable CMAKE_PREFIX_PATH doesn't exist
[0.191s] WARNING:colcon.colcon_ros.prefix_path.catkin:The path '/home/park/Documents/ws_limo_humble/install/limo_car' in the environment variable CMAKE_PREFIX_PATH doesn't exist
Starting >>> limo_car
Starting >>> limo_description
Finished <<< limo_description [0.76s]
Finished <<< limo_car [0.81s]
Finished <<< limo_msgs [3.06s]
Starting >>> limo_base
--- stderr: limo_base
/home/park/Documents/ws_limo_humble/src/limo_ros2/limo_base/src/limo_driver.cpp: In member function 'void AgileX::LimoDriver::parseFrame(const AgileX::LimoFrame&)':
298 |         int32_t left_wheel_odom = (frame.data[3] & 0xff) | (frame.data[2] << 8) |
      |                                     ^~~~~~
/home/park/Documents/ws_limo_humble/src/limo_ros2/limo_base/src/limo_driver.cpp:300:21: warning: unused variable 'right_wheel_odom' [-Wunused-variable]
300 |         int32_t right_wheel_odom = (frame.data[7] & 0xff) | (frame.data[6] << 8) |
      |                                     ^~~~~~
/home/park/Documents/ws_limo_humble/src/limo_ros2/limo_base/src/limo_driver.cpp: In member function 'void AgileX::LimoDriver::publishIMUData(double)':
489 |         double present_theta = imu_data._yaw;
      |                                     ^~~~~~
/home/park/Documents/ws_limo_humble/src/limo_ros2/limo_base/src/limo_driver.cpp:490:16: warning: unused variable 'last_theta_' [-Wunused-variable]
490 |         double last_theta_ = imu_data._yaw;
      |                                     ^~~~~~
---
Finished <<< limo_base [9.22s]

Summary: 4 packages finished [12.4s]
1 package had stderr output: limo_base
root@park:/home/park/Documents/ws_limo_humble# source install/setup.bash
```

- Gazebo 실행

```
ros2 launch limo_car roboracer_4diff_sim.launch.py
```



- teleop

```
# 다른 터미널
docker exec -it limo_gazebo_humble /bin/bash

source /opt/ros/humble/setup.bash

source install/setup.bash

ros2 run teleop_twist_keyboard teleop_twist_keyboard \
  --ros-args -r cmd_vel:=/cmd_vel
```